Chapter 4

Previous experiences

4.1 Leveraging Consistency Models for Open Vocabulary 2D/3D Semantic Segmentation

One of the challenges identified in Chapter 3 is the sensitivity of diffusion-based segmentation pipelines to the denoising timestamp, a parameter with no clear method of selection. This work directly targets that limitation.

4.1.1 Abstract

Text-to-image generative models have gained increasing attention, in the context of open-vocabulary semantic segmentation, due to their ability to capture low-level details in images. Such architectures are competitive with previous standards, based on contrastive methods, whose objective function focuses on high-level, discriminative image features, lacking the details needed for accurate semantic segmentation masks. Among those generative models, text-to-image Diffusion models have been leveraged for open-vocabulary semantic segmentation, without the need for additional training.

When using the internal representations of Diffusion models, for computing segmentation maps, it is required to choose a value for the diffusion timestamp. Because there is no clear method of choosing the denoising timestamp, we propose replacing the Diffusion model backbone with a Consistency model, whose consistency loss objective smoothens out the variation in segmentation accuracy, for different values of denoising timestamp.

4.1.2 Motivation

DiffSegmenter [2] uses the attention maps of StableDiffusion [25] to perform open vocabulary, training-free semantic segmentation. The inference method, for the underlying diffusion model, is DDIM [26]. Figure 4.1 illustrates the components of the DiffSegmenter pipeline and the possibility of adopting more recent models, from the diffusion model family, with the goal of improving accuracy and inference speed of the pipeline. One instance of more recent diffusion models is the Consistency model [24].

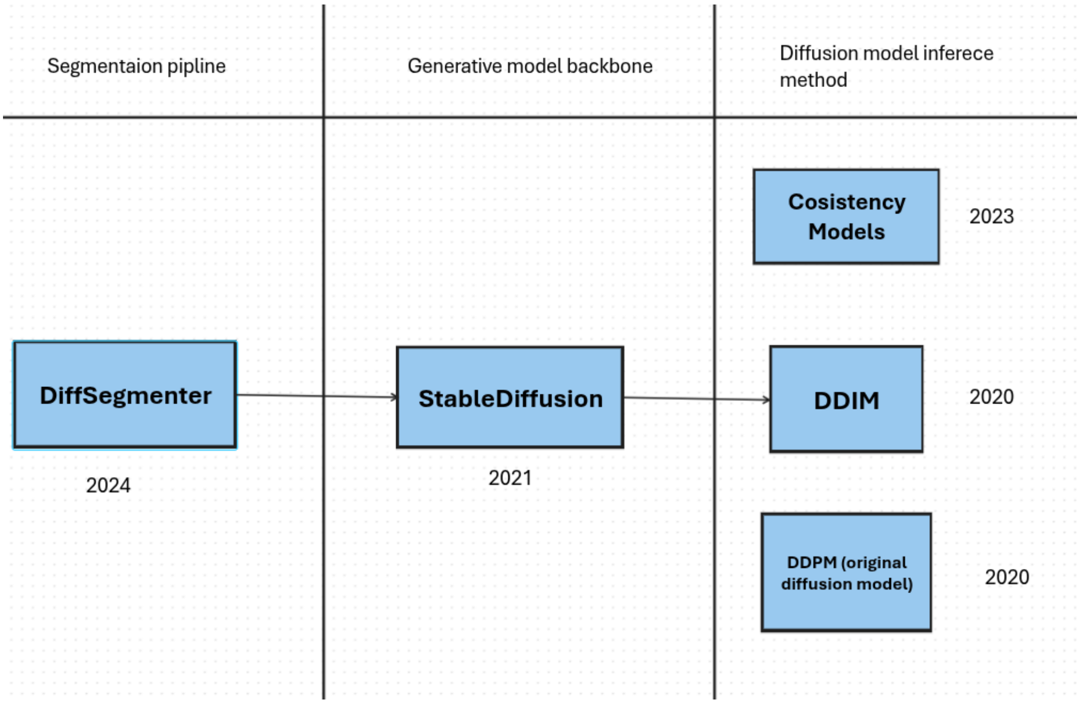

Figure 4.1: Overview of DiffSegmenter components

Authors of DiffSegmenter tested different values for the diffusion model inference parameter: denoising timestamp t, and concluded that averaging across multiple values achieves the best performance 4.2. Thus, the goal of this work is to eliminate the need for choosing a certain value for the denoising timestamp, since there is no clear method of choosing it.

Method	t=1	t=50	t=100	t=150	**Avg.**
mIoU	69.10	69.94	70.30	69.69	70.49

Figure 4.2: Experiments for values of denoising timestamp

Consistency Models [24] use the consistency loss term, giving the model the ability to perform image generation in much fewer steps, at a small cost in image quality. Since the image generation process is consistent across different noise level, using Consistency Models in the DiffSegmenter pipeline has the potential to eliminate the need to aggregate results from multiple generation steps. Highlighted in table 4.3, the text-to-image pipeline LCM, based on Consistency models, outperforms previous methods, for cases when fewer inference steps are used. The performance gains measure the quality of the generated images.

Model (512 × 512) Reso	FID ↓				CLIP Score ↑			
	1 Step	2 Steps	4 Steps	8 Steps	1 Steps	2 Steps	4 Steps	8 Steps
DDIM (Song et al., 2020a)	183.29	81.05	22.38	13.83	6.03	14.13	25.89	29.29
DPM (Lu et al., 2022a)	185.78	72.81	18.53	12.24	6.35	15.10	26.64	29.54
DPM++ (Lu et al., 2022b)	185.78	72.81	18.43	12.20	6.35	15.10	26.64	**29.55**
Guided-Distill (Meng et al., 2023)	108.21	33.25	15.12	13.89	12.08	22.71	27.25	28.17
LCM (Ours)	**35.36**	**13.31**	**11.10**	**11.84**	**24.14**	**27.83**	**28.69**	28.84

Table 1: Quantitative results with $\omega = 8$ at 512×512 resolution. LCM significantly surpasses baselines in the 1-4 step region on LAION-Aesthetic-6+ dataset. For LCM, DDIM-Solver is used with a skipping step of $k = 20$.

Figure 4.3: LCM results

Replacing the underlying Diffusion Model with Consistency Models introduces the need to distill StableDiffusion into a text-conditioned Concistency Model. The following paper introduces text-conditioned Consistency Models: [27].

The work of [28] mentions the ideea of replacing the DDIM backbone with a more recent architecture, in the context of object detection.

4.1.3 Results

The proposed model, LCMSegmenter, aims to improve segmentation results by replacing the text-to-image diffusion model from DiffSegmenter with a Latent Conisistency Model. The text-to-image pipeline used in DiffSegmenter is Stable-Diffusion v5 [29] and the replacement pipeline is LCM_Dreamshaper_v7 [27]. The details of the two architectures are presented below.

	Stable-Diffusion v1-5	LCM_Dreamshaper_v7
Training data	Trained on LAION-2B and subsets.	Distilled from Dreamshaper v7 with 4,000 training interations. Dreamshaper is a fine-tune of Stable-Diffusion v1-5
Parameters count	943,174,827	943,256,747

The presented pipelines have approximately the same number of parameters and the same data was used for their training. Thus, their direct comparison for open vocabulary semantic segmentation is relevant.

The benchmark used in this section is Pascal VOC 2012 [30], a data split of of 1449 samples with one or more classes. Table below 4.4 shows the mean intersection over union for DiffSegmenter [2] (green columns) and the proposed model LCMSegmenter (yellow columns) (Latent Consistency Model Segmenter). The threshold is chosen on 50% of the data, the other 50% is used for computing test mean intersection over union.

timestamp	threshold	val mIoU	test mIoU	test std(mIoU)	threshold	val mIoU	test mIoU	test std(mIoU)
0.01	0.41	58.51	61.58	21.93	0.5	57.51	60.48	21.24
0.1	0.44	59.03	61.05	21.64	0.5	57.68	60.7	21.1
0.2	0.47	58.06	60.53	21.63	0.5	57.69	60.87	21.0
0.3	0.49	56.62	59.58	21.29	0.5	57.64	60.98	20.95
0.4	0.51	54.65	57.32	20.62	0.5	57.59	61.04	20.94
0.5	0.54	50.9	52.55	19.96	0.51	57.52	60.93	20.89
0.6	0.55	45.08	46.41	19.43	0.51	57.44	60.97	20.87
0.7	0.58	38.76	40.51	17.67	0.51	57.38	60.99	20.86
0.8	0.61	33.23	34.63	15.18	0.51	57.31	61.01	20.83

Figure 4.4: VOC2012 results

The underlying Diffusion model of DiffSegmenter is trained on 1000 denoising timestamps and the Diffusion model of LCMSegmenter is trained on 50. For a relevant comparison, the first column shows the denoising timestamp used during inference, relative to the total number of timestamps used for the training of the underlying Diffusion model. For example, 0.1 will mean timestamp 100 for DiffSegmenter, but timestamp 5 for LCMSegmenter.

Results show less variation in accuracy for LCMSegmenter, for different configurations of denoising timestamp (columns with red contour). Also, the threshold changes much more for DiffSegmenter, in relation with the timestamp parameter.

Table 4.5 shows results on the Pascal Context dataset [31]. The improved stability of accuracy and threshold selection, for different configurations of denoising timestamp, are observed for this benchmark as well. In this case, 5104 samples were used to compute results, 20% for threshold selection and 80% for testing. The dataset contained masks for 495 different classes.

timestamp	threshold	val mIoU	test mIoU	test std(mIoU)	threshold	val mIoU	test mIoU	test std(mIoU)
0.001	0.14	11.79	13.68	18.6	0.14	11.11	13.47	18.72
0.1	0.05	11.47	13.52	18.45	0.14	11.13	13.64	18.75
0.2	0.13	10.69	13.19	18.36	0.14	11.14	13.7	18.71
0.3	0.13	10.03	12.45	17.83	0.14	11.16	13.74	18.68

Figure 4.5: VOC Context results

4.1.4 Conclusion

The proposed LCMSegmenter shows that replacing the diffusion backbone with a Consistency Model leads to more stable segmentation accuracy across timestamp configurations, addressing a concrete and previously open challenge in generative approaches. The results suggest that more recent generative architectures are viable and beneficial for open vocabulary semantic segmentation, opening the door to further exploration in this direction.

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